

Assignment 2

Assignment Questions

`pwd` prints the current working directory.

`whoami` prints the username of the currently logged-in user.

The semicolon (;) separates commands so `ls` and `date` run sequentially.

When ";" is quoted, it is treated as a literal argument, so `ls` tries to list a file named ";" and `date` is treated as another argument.

`mkdir -p` creates all directories at once, including parent directories.

`ls -R` recursively lists all files and subdirectories under `data`.

The `umask` controls default permissions for newly created files.

After changing the `umask`, the new file had more restrictive permissions.

The hard link still works because it points to the same inode.

The symbolic link breaks because its target file was deleted.

`diff` shows line-by-line differences.

`cmp` reports the first byte where files differ.

Pipelines avoid intermediate files, are faster, and reduce disk I/O.

`fgrep` treats patterns literally, so it is ideal for searching fixed strings containing regex metacharacters.

Commands

```
Last login: Wed Jan 21 05:20:11 2026 from 47.157.70.152
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-47-37:~$ mkdir -p ~/cmps260/linux_assignment/{data,outputs,scripts}
cd ~/cmps260/linux_assignment
pwd
/home/ubuntu/cmps260/linux_assignment
whoami
ubuntu
ubuntu@ip-172-31-47-37:~/cmps260/linux_assignment$ pwd
/home/ubuntu/cmps260/linux_assignment
whoami > outputs/problem0_commands.txt
/home/ubuntu/cmps260/linux_assignment
ubuntu@ip-172-31-47-37:~/cmps260/linux_assignment$ ls ; date
ls ";" date
data outputs scripts
Wed Jan 21 22:33:27 UTC 2026
ls: cannot access ';': No such file or directory
ls: cannot access 'date': No such file or directory
ubuntu@ip-172-31-47-37:~/cmps260/linux_assignment$ man -f ls
man -k shell
man 1 ls
ls: nothing appropriate.
shell: nothing appropriate.
ubuntu@ip-172-31-47-37:~/cmps260/linux_assignment$ ls ; date1{ man -f ls; man -k shell; man 1 ls | head -n 30
; } >> outputs/problem1.txt
-bash: syntax error near unexpected token `}'
ubuntu@ip-172-31-47-37:~/cmps260/linux_assignment$
{ man -f ls; man -k shell; man 1 ls | head -n 30; } >> outputs/problem1.txt
-bash: syntax error near unexpected token `}'
ubuntu@ip-172-31-47-37:~/cmps260/linux_assignment$ mkdir -p ~/cmps260/linux_assignment/{data,outputs,scripts}
{ man -f ls; man -k shell; man 1 ls | head -n 30; } >> outputs/problem1.txt
ubuntu@ip-172-31-47-37:~/cmps260/linux_assignment$ mkdir -p data/projects/{raw,processed} data/{notes,archive}
mkdir -p data/projects/{raw,processed} data/{notes,archive}
ls -R data >> outputs/problem1_lecture2.txt
ls -R data >> outputs/problem1_lecture2.txt
ubuntu@ip-172-31-47-37:~/cmps260/linux_assignment$ cat > data/notes/notes1.txt
This is line one
This line has multiple spaces

This line comes after a blank line
Here is a tab separated line

Another normal sentence

Yet another line
Line nine
Line ten

ubuntu@ip-172-31-47-37:~/cmps260/linux_assignment$ cat >> data/notes/notes1.txt
```

```

ubuntu@ip-172-31-47-37:~/cmps260/linux_assignment$ cat >> data/notes/notes1.txt
Yet another line
line nine
line ten
ubuntu@ip-172-31-47-37:~/cmps260/linux_assignment$ ^[[200~cat -s data/notes/notes1.txt >> outputs/problem1_lecture2.txt
cat: command not found
ubuntu@ip-172-31-47-37:~/cmps260/linux_assignment$ cat -E data/notes/notes1.txt >> outputs/problem1_lecture2.txt
ubuntu@ip-172-31-47-37:~/cmps260/linux_assignment$ cat -A data/notes/notes1.txt >> outputs/problem1_lecture2.txt
ubuntu@ip-172-31-47-37:~/cmps260/linux_assignment$ tac data/notes/notes1.txt >> outputs/problem1_lecture2.txt
ubuntu@ip-172-31-47-37:~/cmps260/linux_assignment$ ~
-bash: /home/ubuntu: Is a directory
ubuntu@ip-172-31-47-37:~/cmps260/linux_assignment$ echo "dash file" > ./-dashfile.txt
cat ./-dashfile.txt
dash file
ubuntu@ip-172-31-47-37:~/cmps260/linux_assignment$ ls -l
ls -lh
ls -ld data
ls -ltr
ls -A
ls -R data
total 16
-rw-rw-r-- 1 ubuntu ubuntu 10 Jan 21 22:41 -dashfile.txt
drwxrwxr-x 5 ubuntu ubuntu 4096 Jan 21 22:38 data
drwxrwxr-x 2 ubuntu ubuntu 4096 Jan 21 22:38 outputs
drwxrwxr-x 2 ubuntu ubuntu 4096 Jan 21 22:32 scripts
total 16K
-rw-rw-r-- 1 ubuntu ubuntu 10 Jan 21 22:41 -dashfile.txt
drwxrwxr-x 5 ubuntu ubuntu 4.0K Jan 21 22:38 data
drwxrwxr-x 2 ubuntu ubuntu 4.0K Jan 21 22:38 outputs
drwxrwxr-x 2 ubuntu ubuntu 4.0K Jan 21 22:32 scripts
drwxrwxr-x 5 ubuntu ubuntu 4096 Jan 21 22:38 data
total 16
drwxrwxr-x 2 ubuntu ubuntu 4096 Jan 21 22:32 scripts
drwxrwxr-x 5 ubuntu ubuntu 4096 Jan 21 22:38 data
drwxrwxr-x 2 ubuntu ubuntu 4096 Jan 21 22:38 outputs
-rw-rw-r-- 1 ubuntu ubuntu 10 Jan 21 22:41 -dashfile.txt
-dashfile.txt data outputs scripts
data:
archive notes projects
data/archive:
data/notes:
notes1.txt

```

```

Tasks: 108 total, 1 running, 107 sleeping, 0 stopped, 0 zombie
Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
 8 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns

10 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri

4925 ubuntu  20  0    12372   5896   3664 R   0.3   0.6   0:00.08 top

 1 root      20  0    22636   13964   9628 S   0.0   1.5   0:04.43 systemd
 2 root      20  0      0      0      0 S   0.0   0.0   0:00.01 kthreadd
 3 root      20  0      0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
 4 root       0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
 5 root       0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
 6 root       0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
 7 root       0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
 8 root       0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
10 root       0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
13 root       0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
14 root      20  0      0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
15 root      20  0      0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
16 root      20  0      0      0      0 S   0.0   0.0   0:00.06 ksoftirqd/0
17 root      20  0      0      0      0 I   0.0   0.0   0:01.02 rcu_sched
18 root      20  0      0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
19 root      20  0      0      0      0 S   0.0   0.0   0:00.00 rcu_exp_gp_kthread_worker
20 root      rt  0      0      0      0 S   0.0   0.0   0:00.30 migration/0
top - 22:46:14 up 18:02,  1 user,  load average: 0.00, 0.00, 0.00
Tasks: 107 total, 1 running, 106 sleeping, 0 stopped, 0 zombie

22 root      20  0      0      0      0 S   0.0   0.0   0:00.00 cpuhp/0

```

```
Charlie
alpha
apples apples
oranges oranges
walnuts grapes
alpha OK
alpha FAIL
bravo 19.99
bravo 100.00
charlie 7.50
alpha products
bravo checkout
charlie OK
alpha OK
ubuntu@ip-172-31-47-37:~/cms260/linux_assignment$ history > outputs/problem2_commands.txt
ubuntu@ip-172-31-47-37:~/cms260/linux_assignment$ grep '^[a-z]' data/regex_lab.txt
egrep '^[a-zA-Z ][a-zA-Z 0-9]*=' data/regex_lab.txt
egrep '\${0-9}+(\.[0-9]{2})?' data/regex_lab.txt
egrep '<[Hh][1-6]>' data/regex_lab.txt
grep: data/regex_lab.txt: No such file or directory
grep: data/regex_lab.txt: No such file or directory
grep: data/regex_lab.txt: No such file or directory
grep: data/regex_lab.txt: No such file or directory
ubuntu@ip-172-31-47-37:~/cms260/linux_assignment$ fgrep 'a*b*' data/regex_lab.txt
grep 'a*b*' data/regex_lab.txt
grep: data/regex_lab.txt: No such file or directory
grep: data/regex_lab.txt: No such file or directory
ubuntu@ip-172-31-47-37:~/cms260/linux_assignment$ tr 'a-z' 'A-Z' < data/regex_lab.txt
tr -s ' ' < data/regex_lab.txt
tr -d '0-9' < data/regex_lab.txt
-bash: data/regex_lab.txt: No such file or directory
-bash: data/regex_lab.txt: No such file or directory
-bash: data/regex_lab.txt: No such file or directory
ubuntu@ip-172-31-47-37:~/cms260/linux_assignment$ sed 's/login/LOGIN/' data/log.txt
sed 's/pay/PAY/g' data/log.txt
sed -n 's/OK/SUCCESS/p' data/log.txt
sed '3,5d' data/log.txt
alpha LOGIN OK
alpha LOGIN FAIL
bravo pay 19.99
bravo pay 100.00
charlie pay 7.50
alpha browse products
bravo browse checkout
charlie LOGIN OK
alpha LOGIN OK
alpha login OK
alpha login FAIL
bravo PAY 19.99
```

```
tr -s ' ' < data/regex_lab.txt
tr -d '0-9' < data/regex_lab.txt
-bash: data/regex_lab.txt: No such file or directory
-bash: data/regex_lab.txt: No such file or directory
-bash: data/regex_lab.txt: No such file or directory
ubuntu@ip-172-31-47-37:~/cmcs260/linux_assignment$ sed 's/login/LOGIN/' data/log.txt
sed 's/pay/PAY/g' data/log.txt
sed -n 's/OK/SUCCESS/p' data/log.txt
sed '3,5d' data/log.txt
alpha LOGIN OK
alpha LOGIN FAIL
bravo pay 19.99
bravo pay 100.00
charlie pay 7.50
alpha browse products
bravo browse checkout
charlie LOGIN OK
alpha LOGIN OK
alpha login OK
alpha login FAIL
bravo PAY 19.99
bravo PAY 100.00
charlie PAY 7.50
alpha browse products
bravo browse checkout
charlie login OK
alpha login OK
alpha login SUCCESS
charlie login SUCCESS
alpha login SUCCESS
alpha login OK
alpha login FAIL
alpha browse products
bravo browse checkout
charlie login OK
alpha login OK
ubuntu@ip-172-31-47-37:~/cmcs260/linux_assignment$ #!/bin/bash
grep '^time' data/regex_lab.txt
tr -d '0-9' < data/regex_lab.txt
sed 's/HTML/WEB/' data/regex_lab.txt
grep: data/regex_lab.txt: No such file or directory
-bash: data/regex_lab.txt: No such file or directory
sed: can't read data/regex_lab.txt: No such file or directory
ubuntu@ip-172-31-47-37:~/cmcs260/linux_assignment$ chmod +x scripts/problem3_regex_tr_sed.sh
bash scripts/problem3_regex_tr_sed.sh > outputs/problem3_lecture4.txt
history > outputs/problem3_commands.txt
chmod: cannot access 'scripts/problem3_regex_tr_sed.sh': No such file or directory
bash: scripts/problem3_regex_tr_sed.sh: No such file or directory
ubuntu@ip-172-31-47-37:~/cmcs260/linux_assignment$
```